PAPER EXPLORATION

# INSIDER THREAT DETECTION BASED ON USERS’ MOUSE MOVEMENTS AND KEYSTROKES BEHAVIOR

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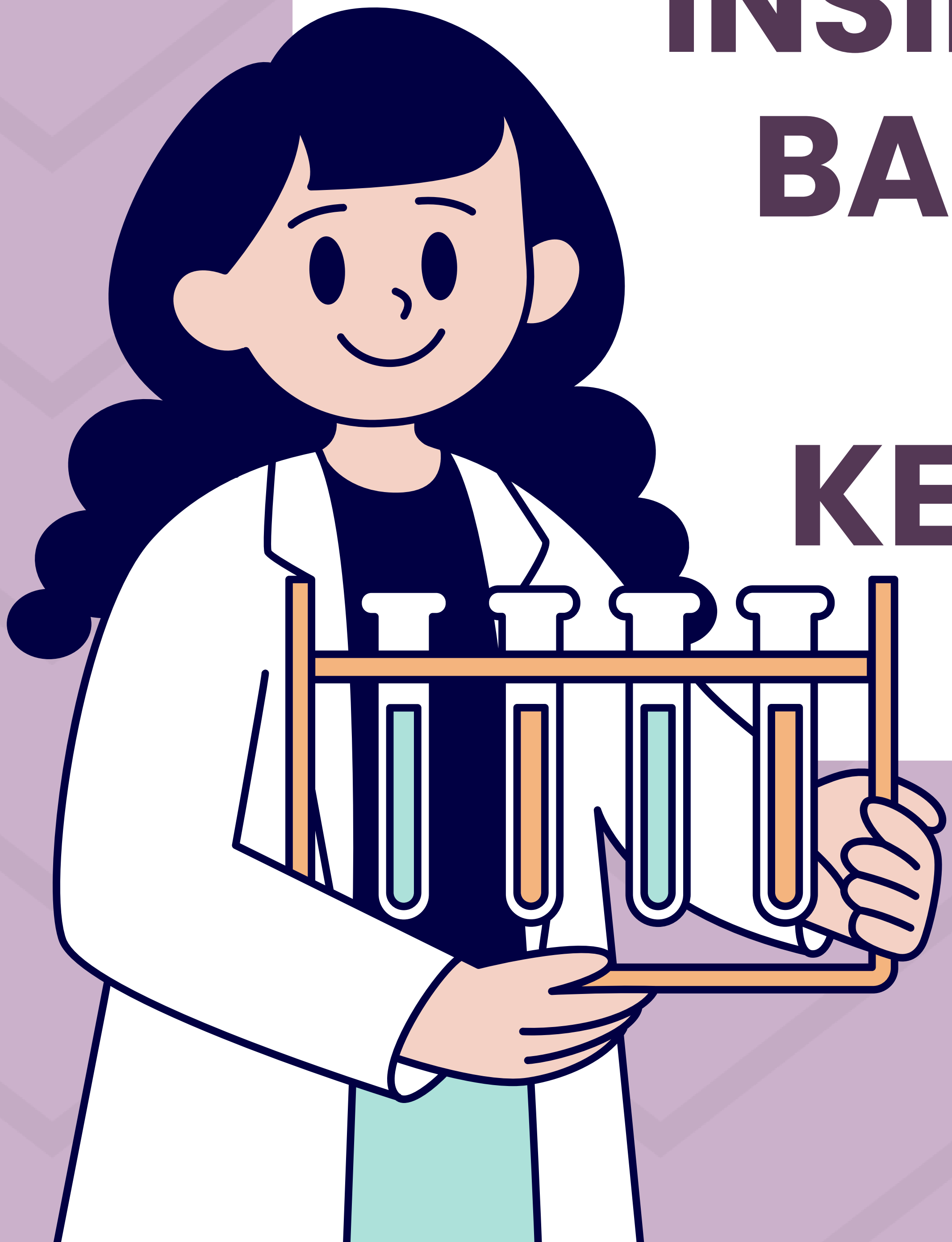


# PROBLEM

- The problem addressed in this paper is the detection of insider threats in organizations.
- Traditional methods of detecting insider threats have limitations, and there is a need for alternative approaches.

# SOLUTION

- The proposed solution by the authors is to analyze mouse movements and keystroke dynamics as indicators of malicious intent.
- By studying the patterns and characteristics of mouse movements and keystrokes, the authors aim to identify and predict insider threats more effectively.

# HISTORY

- The analysis of mouse movements and keystroke dynamics for insider threat detection is a relatively unexplored area, with previous research mainly focusing on authentication and identification purposes

- The research conducted by the authors builds upon their previous work in multi-modal neurophysiological studies of malicious insider threats

# EXPERIMENTAL SETUP:

- **Participants:** The experiment involved a total of 30 participants, including graduate and undergraduate students from the University of North Texas.
- **Controlled Lab Conditions:** The experiments were conducted in a lab room specifically set up to maintain consistent environmental conditions for all tasks and participants.
- **Diverse Backgrounds:** The participants were selected to represent a diverse range of backgrounds, including different levels of programming skills and cybersecurity knowledge.
- **Different Tasks:** The experiment consisted of six different tasks, including both benign daily activities and malicious activities, to simulate real-world scenarios

# DATA COLLECTION AND ANALYSIS:

## MOUSE MOVEMENTS AND KEYSTROKES

The authors recorded and analyzed the participants' mouse movements and keystroke dynamics using software that captured the position, event, and timestamp of mouse movements and keystrokes

## FEATURE EXTRACTION

Various features were extracted from the mouse movements and keystrokes, such as mouse movement speed, distance traveled, left click duration, and keystroke duration

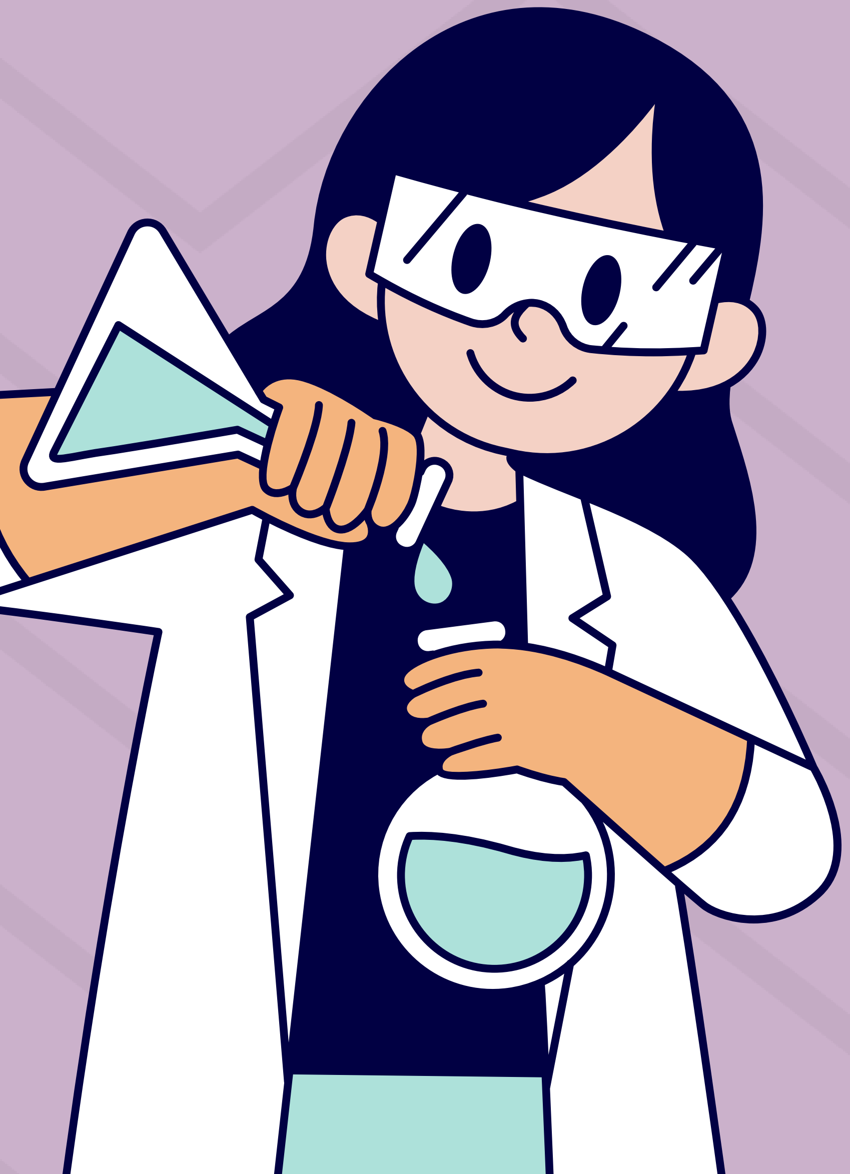
## CLASSIFICATION AND STATISTICAL ANALYSIS

Classification algorithms and statistical measures were used to evaluate the extracted features and analyze the differences in behavior patterns between benign and malicious activities

# CLASSIFIERS RESULTS

- **Average Detection Accuracy:** The researchers used several classification algorithms, including Support Vector Machine (SVM), K-nearest-neighbors (k-NN), Random Forests, and Bagging Predictors. The average detection accuracy is between 67.5% and 72.5% in detecting malicious activities based on mouse movements and keystroke dynamics.

- **Use of Statistical Analysis:** The researchers conducted statistical analysis in addition to using classifiers to investigate the significance of different features and gain a deeper understanding of the behaviour patterns associated with malicious activities.



# STATISTICAL FINDINGS:

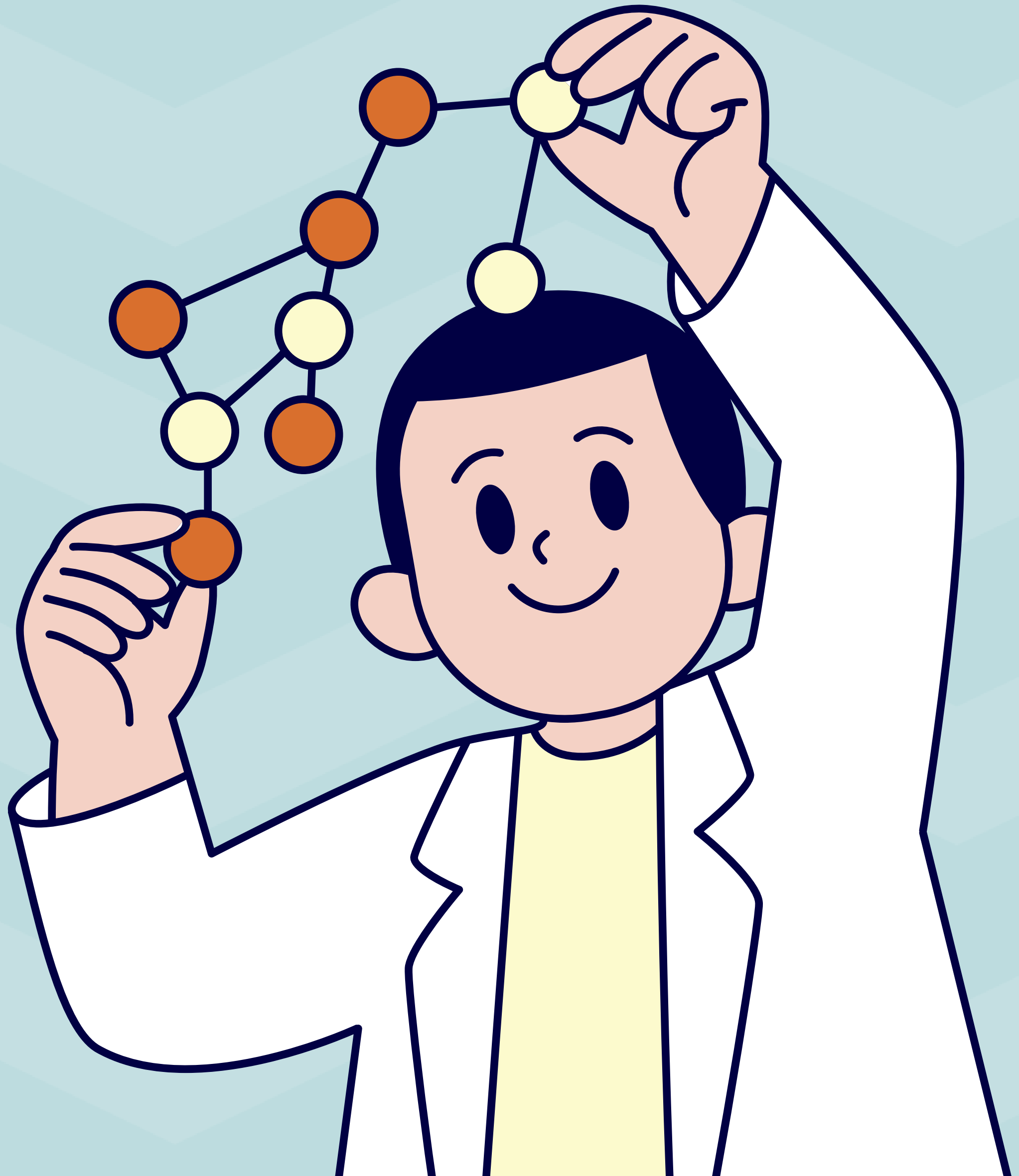
| Feature | Benign Mean | Benign under Stress Mean | Malicious Mean | p-value (Malicious and Benign task) | p-value (Malicious and under stress) |
|---|---|---|---|---|---|
| Mouse Movement Speed | 0.75 | 0.82 | 0.97 | 0.0000316 | 0.034 |
| Mouse Travel Distance | 145.93 | 146.57 | 181.13 | 0.000832 | 0.00114 |
| Left Mouse Click Duration | 167.63 | 226.69 | 293.01 | 0.000015 | 0.0451 |
| Keystroke Duration | 218.52 | 506.65 | 469.32 | 0.0385 | 0.418571 |

- Mouse Movement Speed: Participants exhibited higher mouse movement speeds during malicious tasks compared to benign tasks even under stress
- Mouse Travel Distance: Participants covered greater distances with their mouse movements during malicious tasks compared to normal pattern even under stress.
- Left Mouse Click Duration: Participants had longer-lasting left mouse clicks during malicious tasks compared to benign tasks
- Keystroke Duration: Participants had longer keystroke durations during malicious tasks compared to benign tasks. User under stress and malicious may have similarity

# FUTURE WORKS

- Integration with Multi-Modal Framework: In future work, the extracted mouse movements and keystroke features can be integrated into a multi-modal framework that incorporates other behavioural indicators. This integration can enhance the accuracy and effectiveness of insider threat detection systems.

- Real-World Scenario Testing: The current study utilized controlled experiments to simulate different tasks and scenarios. Future work should involve testing the approach in real-world settings to evaluate its performance and generalizability in detecting insider threats in actual organizational environments.



# TAKEWAYS

- Mouse movements and keystroke dynamics can be valuable behavioural indicators for detecting threats.
- Statistical analysis can complement classifier results by providing insights into the significance of different features and the variability among participants
- The study emphasizes the importance of investigating and selecting informative features.
- Further research is needed to refine feature extraction techniques, explore advanced classification algorithms, and test the approach in real-world scenarios to improve the accuracy and efficiency of threat detection.

# Thanks!

Any questions?